

What it does

Creates exploded-view maps from polygon data. Regions are separated for display while each polygon is moved by rigid translation, so its internal shape is preserved.

Pick a workflow

Any projected sf	<code>explode_sf()</code>
Lookup table	<code>explode_sf_with_lookup()</code>
US state data	<code>explode_state()</code>
National/grouped layout	<code>explode_grouped()</code>
Interactive map	<code>focus_map()</code>
Manual refinement	<code>as_dragmapr_state()</code>

Quick start

```
library(explodemap)

layout <- explode_sf(my_sf,
  region_col = "region",
  plot = FALSE)
plot(layout, "both")
```

Input checklist

- Polygon or multipolygon sf
- Projected CRS, usually metres
- Group column with at least two groups
- No empty geometries
- Stable IDs are recommended

Prepare and validate

```
validate_explodemap_input(
  my_sf, group_col = "region",
  id_col = "geoid", label_col = "name")

prepare_explodemap_input(
  my_sf, id_col = "geoid",
  label_col = "name",
  group_col = "region",
```

```
target_crs = 5070,
target.vertices = 350000)
```

Helpful app primitives

```
count_geometry_vertices()
simplify_to_vertex_budget()
assign_spatial_groups()
group_palette()
explodemap_fingerprint()
```

Grouped layouts

```
explode_grouped(
  my_sf, region_col = "region",
  mode = "auto_collision",
  plot = FALSE)
```

Anchor modes

auto	radial placement
auto_collision	nudged to reduce overlap
manual	user-supplied anchors

Diagnostics and search

```
report <- diagnose_layout(layout,
  label_col = "name")
plot(report)

better <- optimize_grouped_layout(
  my_sf, region_col = "region",
  label_col = "name")
```

Interactive focus map

```
focus_map(
  layout, label_col = "name",
  group_col = "region",
  group_palette = group_palette(groups),
  show_group_labels = TRUE)
```

Shiny

```
focusmapOutput("map")
```

```
renderFocusmap({
  focus_map(layout, label_col = "name")
})
```

Manual refinement with dragmapr

```
library(dragmapr)

state <- as_dragmapr_state(layout)
dragmapr_edit(layout, state = state)

focus_map(layout, state = state)

render_dragged_map(
  layout$sf_grouped,
  region_col = "region", state = state)
```

Common methods

<code>print(x)</code>	quick stats
<code>summary(x)</code>	detailed stats
<code>plot(x)</code>	exploded map
<code>plot(x, "both")</code>	original + exploded
<code>calibration_row(x)</code>	one-row summary

State and HHS helpers

HHS reference maps

```
hhs_regions()   hhs_palette()
hhs_display_offsets() hhs_focus_map()
```

Pipeline Studio

```
shiny::runApp(system.file(
  "shiny/pipeline-studio",
  package = "explodemap"))
```

Remember

- Use `plot = FALSE` in scripts and apps.
- Use `quiet = TRUE` inside Shiny reactives.
- Diagnostics find problems; composition is still editorial.
- Use `dragmapr` when a map needs human placement.